

6.8 Proving Triangle Similarity – Part 1

Lesson Introduction

 Benchmark	MA.912.GR.1.2 Prove triangle congruence or similarity using Side-Side-Side, Side-Angle-Side, Angle-Side-Angle, Angle-Angle-Side, Angle-Angle, and Hypotenuse-Leg.		 Learning Target	I can use triangle similarity to prove triangles are similar.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- How do you prove two triangles are similar? - Why isn't there an AAA, AAS, or an ASA similarity theorem?			
 Lesson Narrative	This lesson brings together the information in Lessons 6 and 7 to write formal proofs to conclude that two triangles are similar. Students see that sometimes proofs can be written using different means of justification. Students also explore why there is no AAA, AAS, or ASA similarity theorem.			
 Component Summary	6.8.1 Warm-Up (~5 min.)	Writing equivalent ratios for triangular garden (<i>SE p. 332</i>)	6.8.3 Guided Instruction (15-20 min.)	Proving triangles similar using a two-column proof (<i>SE p. 334</i>)
	6.8.2 Collaboration (10 min.)	Writing similarity statements and identifying proportional and congruent parts (<i>SE p. 333</i>)	6.8.4 Try It (10 min.)	Practice proving triangles similar using a two-column proof (<i>SE p. 336</i>)
	6.8.5 Wrap-Up (~5 min.)	Ticket Out the Door completing two-column proof (<i>SE p. 337</i>)		
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Similarity - Congruent - Proportional		(Optional) Patty Paper (Optional) Rulers (Optional) Protractors	Consider providing a bank of statements and reasons for students to select from when they are completing two-column proofs.

 Teacher Tips	Common Misconceptions		Things to Consider	
	- Students may write similarity statements incorrectly by not having the corresponding vertices in the same order. - Students may mistake naming triangles with similarity statements where the order of corresponding parts is necessary.		- Students may use transitive or substitution properties of equality interchangeably. Clarify your expectations with your students. 6.8.4 Try It is the first place that the Side-Side-Side Similarity Theorem is used. This was done to help students transfer learning. Use observational data to determine whether to make the questions in 6.8.4 Try It part of 6.8.3 Guided Instruction. If doing this, then consider how to monitor class time. - If time is an issue, then consider assigning 6.8.4 Try It for homework.	
	Literacy and Language Routines		Mathematical Thinking and Reasoning (MTR) Standard	
	Think-Pair-Share 6.8.3 Guided Instruction is a great place to offer students an opportunity to work together to process their thinking. This is where students will be able to share different strategies to reach the same conclusion. Students should first work alone before pairing with others. Then have students share with the whole class. This move builds students' confidence before sharing with the entire class.		MA.K12.MTR.4.1: Discussion and Reflection 6.8.2 Collaboration offers students many opportunities to work together and discuss key mathematical concepts and practices for this lesson. The partnered approach continues for 6.8.3 Guided Instruction. While students complete question 1d look for opportunities to develop students' ability to justify methods and compare their response to the responses of their peers. Offer praise and support to students, especially to those whose confidence level may not be as high as other students.	
 Differentiate Instruction	Students who struggle with writing proofs will benefit from the scaffolding provided in the proofs of this lesson. Scaffolding is seen in the two-column proofs with the inclusion of some of the statements and reasons. Some students are ready for a challenge and can be given the proofs on separate paper without the scaffolds built in.			
Lesson Notes:				

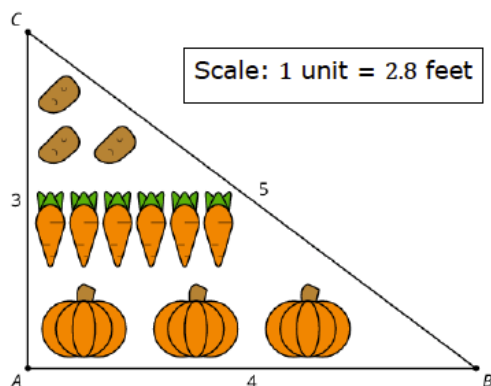
6.8.1 Warm-Up: Vegetable Garden

Component Overview: Students write equivalent ratios after making observations about a vegetable garden.



6.8.1 Warm-Up: Vegetable Garden

1. $\triangle ABC$ is used to create a plan for a vegetable garden.



- a. What do you notice about the plan?

Answers will vary. I notice the plan is a triangle. I notice that they could plant more by changing how they are arranged.

- b. Write at least three equivalent ratios or equations using lengths from both the diagram and the full-size garden.

Answers will vary.

$$\frac{3}{8.4} = \frac{5}{14}; \frac{4}{11.2} = \frac{5}{14}; \frac{11.2}{4} = \frac{8.4}{3}$$



As time allows, have a variety of students share their responses and reasoning with the whole class. Consider asking students if they have any personal experience planting gardens.

6.8.2 Collaboration: Proving Triangle Similarity

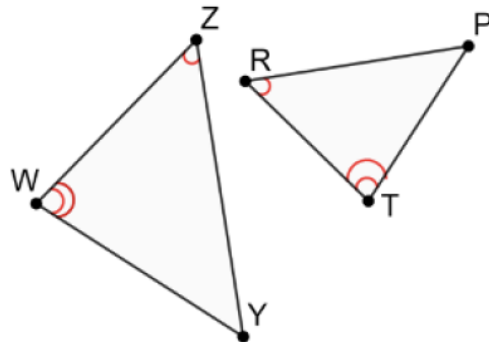
Component Overview: After marking congruent parts on two triangles, students write a similarity statement and identify other proportional and congruent parts.



6.8.2 Collaboration: Proving Triangle Similarity

With your partner, answer the following questions.

1. $\triangle WZY$ and $\triangle RPT$ are shown. $\angle WZY \cong \angle TRP$ and $\angle ZWY \cong \angle RTP$.



- Mark the figure using the given information.
- What similarity theorem can be applied to show that the triangles are similar?
Angle-Angle Similarity
- Write the similarity statement.
 $\triangle WZY \sim \triangle RPT$
- Complete each.

$$\frac{WZ}{TR} = \frac{YW}{PT} \quad \frac{ZY}{RP} = \frac{ZW}{RT} \quad \angle WYZ \cong \angle TPR$$



For students who need support in understanding similarity, consider providing these students with patty paper, rulers, and protractors to further investigate the two triangles.



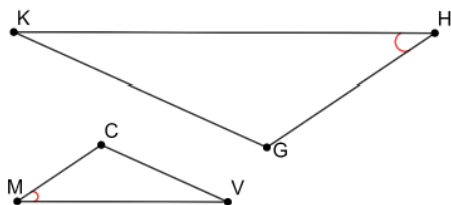
When writing a similarity statement students may use the order of the vertices given in the stem. Help students understand that when just naming triangles order of correspondence is not necessary, but when writing congruence or similarity statements it is necessary.



Why is it possible for your similarity statement to not match your partner's similarity statement and for both to still be correct?

6.8.2 Collaboration: Proving Triangle Similarity (continued)

2. $\triangle KHG$ and $\triangle MVC$ are shown. $\angle KHG \cong \angle VMC$ and $\frac{KH}{VM} = \frac{GH}{CM}$.



- Mark the figure using the given information.
- What similarity theorem can be applied to show that the triangles are similar?
Side-Angle-Side Similarity Theorem
- Write the similarity statement.
 $\triangle KHG \sim \triangle MVC$
- Complete each.

$$\frac{KH}{VM} = \frac{GH}{CM} \quad \angle KHG \cong \angle VMC \quad \angle KGH \cong \angle CVM$$



How can these proportional statements be rewritten and still be true?



Students should regularly be encouraged to develop and share models and tools being used to identify the correct theorem throughout this lesson. Students have many opportunities to work with partners and many tools to pick from. A well-organized “tool box” may help them approach each question with confidence.

6.8.3 Guided Instruction: Proving Triangle Similarity

Component Overview: Students begin reasoning through a two-column proof with a partner, discussing why different methods of proving the conclusion can be used. Students discuss why proving similarity does not need all three angles and how this connects to there being no ASA or AAS similarity theorems. The component concludes with writing a two-column proof to prove the similarity of two triangles.



6.8.3 Guided Instruction: Proving Triangle Similarity

1. $\triangle GTA$ and $\triangle TER$ are shown, and $\overline{GA} \parallel \overline{ER}$.

- a. With your partner, discuss what conclusions can be drawn using the given information.

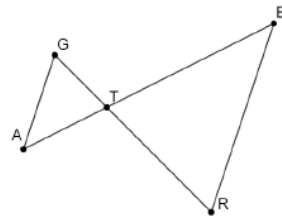
Summarize your discussion and explain how you reached each conclusion.

Answers will vary. We concluded that $\angle AGT \cong \angle ERT$ by alternate interior angles formed by parallel segments $\overline{GA}$ and $\overline{ER}$ and transversal $\overline{GR}$. We concluded that $\angle GAT \cong \angle RET$ by alternate interior angles formed by parallel segments $\overline{GA}$ and $\overline{ER}$ and transversal $\overline{AE}$. We concluded that $\angle GTA \cong \angle RTE$ by the vertical angles theorem and $\triangle GTA \sim \triangle RTE$ by AA similarity.

- b. Which similarity postulates should be used to prove $\triangle GTA \sim \triangle RTE$?

Angle-Angle Similarity

- c. Complete the two-column proof. *Answers may vary.*



Statement	Reason
1. $\overline{GA} \parallel \overline{ER}$	1. <i>Given</i>
2. $\angle AGT \cong \angle ERT$	2. If two parallel lines are intercepted by a transversal, then alternate interior angles are congruent.
3. $\angle GAT \cong \angle RET$	3. If two parallel lines are intercepted by a transversal, then alternate interior angles are congruent.
4. $\triangle GTA \sim \triangle RTE$	4. <i>Angle-Angle Similarity</i>

- d. Ask two classmates about their proof. Record their step 3 statement and reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Answers will vary.



Consider allowing students to work with their partners for all of questions 1a-1e before bringing them together for whole group discussion. This will give students an opportunity to refine their thinking before sharing with the whole group.



Collect and Display

Consider utilizing this language routing when discussing question 1a with the whole class. Direct students to justify their conclusions using formal mathematical language. While the students share their conclusions, display their responses for all to see. Probe students to include which transversal results in congruent alternate interior angles.

6.8.3 Guided Instruction: Proving Triangle Similarity (continued)

Step 3 Statement	Step 3 Reason	Initials

- e. Discuss with your partner the two different ways that $\triangle GTA \sim \triangle RTE$ can be proved.

One method is stating that $\angle GAT \cong \angle TER$ and $\angle AGT \cong \angle ERT$ using alternate interior angles and then stating the triangles similar by AA similarity. Another method is stating either $\angle GAT \cong \angle TER$ or $\angle AGT \cong \angle ERT$ using alternate interior angles and stating that $\angle ATG \cong \angle RTE$ by vertical angles and then stating the triangles similar by AA similarity.

- f. Explain why you do not need to state three congruent angles in your two-column proof.
If two angles in a triangle are congruent to two angles in another triangle then the third must also be congruent. Therefore, only two angles are needed to be shown congruent for the triangles to be similar.
- g. Arian and Isabella were discussing why there is no Angle-Angle-Angle Similarity Theorem.

Arian	Isabella
I think once you know two pairs of angles in a triangle are congruent, then you know the last pair are congruent since the sum of all of the angles of a triangle are 180° .	I think once you know two pairs of angles in a triangle are congruent, then you know all of the sides are proportional.

Discuss with your partner why Arian and Isabella's arguments could also be applied to show why Angle-Side-Angle similarity and Angle-Angle-Side similarity are not needed to prove similarity. Summarize your discussion.

Having two pairs of congruent angles is sufficient to prove triangles similar. It is not necessary to have any of the side measures in this case to prove similarity.



Monitor while students are discussing and sharing their step 3 statements and reasons. Look for opportunities to develop students' ability to justify methods and compare their responses to the responses of their peers. Offer praise and support to students, especially to those whose confidence level may not be as high as other students.



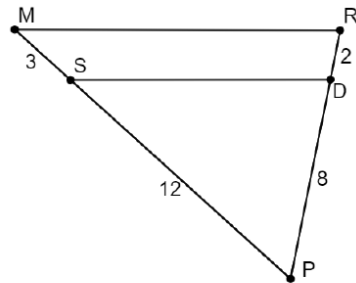
Question 1e is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine students' thinking to highlight or their misconceptions to address.

6.8.3 Guided Instruction: Proving Triangle Similarity (continued)

2. $\triangle SPD$ and $\triangle MPR$ are shown.

Given: $MS = 3$, $SP = 12$, $RD = 2$, and $DP = 8$

Prove: $\triangle SPD \sim \triangle MRP$



Statement	Reason
1. $MS = 3$, $SP = 12$, $RD = 2$, and $DP = 8$	1. <i>Given</i>
2. $MS + SP = PM$ $RD + DP = RP$	2. <i>Segment Addition Postulate</i>
3. $PM = 15$ $RP = 10$	3. Substitution Property of Equality
4. $\frac{PS}{PM} = \frac{12}{15}$ $\frac{PD}{PR} = \frac{8}{10}$	4. <i>Division Property of Equality</i>
5. $\frac{PS}{PM} = \frac{PD}{PR}$	5. <i>Substitution Property of Equality</i>
6. $\angle P \cong \angle P$	6. <i>Reflexive Property of Equality</i>
7. $\triangle SPD \sim \triangle MRP$	7. <i>Side-Angle-Side Similarity</i>



Consider implementing a silent board game for this proof. Display the proof at the front of the room and have students volunteer to go to the board to complete one line of the proof. Students do not talk during this activity. This game allows students who don't see all the reasons in the entire proof an entry-point, as they may know some of the reasons.

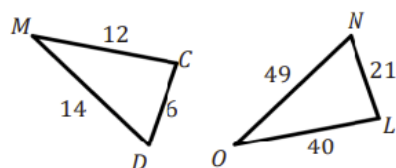
6.8.4 Try It: Triangle Similarity

Component Overview: Students consider reasoning to determine if two triangles are similar given only the side lengths before completing a two-column proof proving that two triangles are similar.



6.8.4 Try It: Triangle Similarity

1. Amanda and Tyler are determining if $\triangle MCD \sim \triangle OLN$ is true. Their reasonings are shown.



Amanda's Answer	Tyler's Answer
No, the triangles are not similar because the corresponding sides are not proportional.	Yes, the triangles are similar because the corresponding sides are proportional.

Who do you agree with? Explain your reasoning.

I agree with Amanda because when I divided the side lengths of $\triangle MCD$ by the corresponding side lengths of $\triangle OLN$, they are not equal.



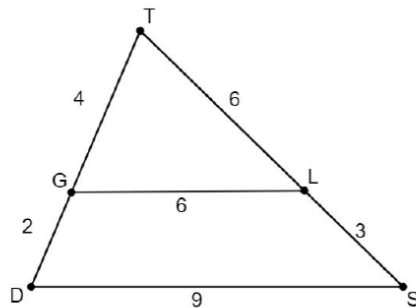
During this component is the first time students will use SSS similarity in the lesson. Listen as students work through question 1 with their partner to identify misconceptions about what's needed to prove similarity given the side lengths before students move on to the proof in question 2.

6.8.4 Try It: Triangle Similarity (continued)

2. $\triangle TLG$ and $\triangle TSD$ are shown.

Given: $TG = 4$, $GD = 2$, $TL = 6$, $LS = 3$, $GL = 6$, and $DS = 9$

Prove: $\triangle TLG \sim \triangle TSD$



Statement	Reason
1. $TG = 4$, $GD = 2$, $TL = 6$, $LS = 3$, $GL = 6$, and $DS = 9$	1. <i>Given</i>
2. $TG + GD = TD$ $TL + LS = TS$	2. <i>Segment Addition Postulate</i>
3. $TD = 6$, $TS = 9$	3. Substitution Property of Equality
4. $\frac{TG}{TD} = \frac{4}{6}$, $\frac{TL}{TS} = \frac{6}{9}$, $\frac{GL}{DS} = \frac{6}{9}$	4. <i>Division Property of Equality</i>
5. $\frac{TG}{TD} = \frac{TL}{TS} = \frac{GL}{DS}$	5. <i>Substitution Property of Equality</i>
6. $\triangle TLG \sim \triangle TSD$	6. <i>Side-Side-Side similarity</i>



Students may have a hard time knowing what reasons and statements to fill in. Consider providing an answer choice bank for students to choose from for this component.



For Reason 3, students may also use the substitution property of equality.



Question 2 is the first place where students encounter the Side-Side-Side Similarity Theorem used in a proof. This was done to help them transfer learning. If your students are not quite ready for such a challenge, then consider making this question more guided.

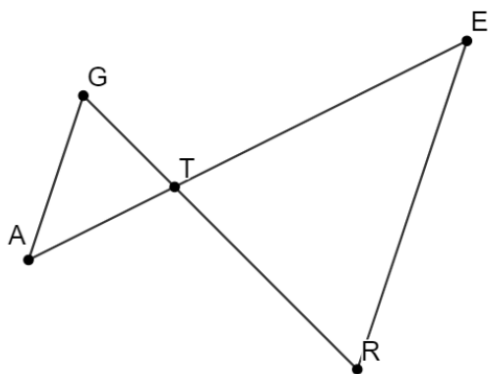
6.8.5 Wrap-Up: Ticket Out the Door

Component Overview: Students fill in statements and reasons in a two-column proof to demonstrate that the given triangles are similar.



6.8.5 Wrap-Up: Ticket Out the Door

1. $\triangle GTA$ and $\triangle TER$ are shown.



Given: $TA = \frac{2}{3}TE$ and $TG = \frac{2}{3}RT$

Prove: $\triangle GTA \sim \triangle RTE$

Complete the two-column proof.

Statement	Reason
1. $TA = \frac{2}{3}TE$ and $TG = \frac{2}{3}RT$	1. Given
2. $\frac{TA}{TE} = \frac{2}{3}$, $\frac{TG}{RT} = \frac{2}{3}$	2. Division Property of Equality
3. $\frac{TA}{TE} = \frac{TG}{RT}$	3. Transitive Property of Equality
4. $\angle GTA \cong \angle RTE$	4. Vertical angles are congruent
5. $\triangle GTA \sim \triangle RTE$	5. Side-Angle-Side Similarity



Consider using the responses here to determine remediation needs on proving triangles are similar.